

Sunrise Dental Clinic System — Assessment Completion

Summary

Module: CIS6003 Advanced Programming (WRIT1) | **Student:** BSc SE – CIS-6003 – 20374265 **Repository:** <https://github.com/TharusanK23/sunrise-dental-clinic> (public, branches `dev` and `release`, both green on CI) **Prepared:** 24 August 2026

This document confirms, task by task against the assignment brief, what has been completed for the Sunrise Dental Clinic System coursework, lists the two most recent changes applied (staff credential rotation and explicit JWT-based authorization), and gives the exact file/URL locations for every piece of evidence referenced.

1. Two most recent changes applied

1.1 Staff credential rotated

The previously-seeded STAFF account has been replaced everywhere:

	Old	New
Username	nadeesha	kirisha
Password	Nadeesha@123	Kirisha@123
Full name	Nadeesha Wickramasinghe	Kirisha N
Email	nadeesha@sunrisedentalclinic.lk	kirisha@sunrisedentalclinic.lk

Updated in: `database/schema.sql` (a genuine, freshly-generated BCrypt hash

- never a placeholder string), every backend test file that referenced the old username, `frontend/index.html`'s `login-hint`, `README.md`, `docs/SETUP.md`, `docs/ASSIGNMENT_REPORT.md`, `testing/TEST_CASES.md`, `testing/postman/SunriseDentalClinic.postman_collection.json`, and `diagrams/sequence-02-register-appointment.mmd` (re-rendered). The admin account (`admin` / `admin123`) is unchanged.

1.2 Explicit JWT-based authorization added

Authorization was already JWT-backed internally, but the token was only ever delivered as an invisible `HttpOnly` cookie. Login now **also** returns the signed token directly in the response body, and this is now documented and demonstrated end-to-end:

- `LoginResponse` gained a `token` field; `AuthController.login()` returns it alongside the existing `dc_token` cookie.
- `JwtAuthenticationFilter` (unchanged - it already supported this) checks the cookie first, then falls back to an `Authorization: Bearer <token>` header - so either mechanism authenticates identically.
- A new `OpenApiConfig` registers a `bearerAuth` security scheme, so Swagger UI's **Authorize** padlock now works: paste the `token` from a login response and every "Try it out" call carries it automatically.
- `testing/postman/SunriseDentalClinic.postman_collection.json`: the Login requests now run a test script that captures `token` into a collection variable, and the whole collection's inherited auth is set to `Authorization: Bearer {{token}}`, alongside Postman's normal cookie jar.
- A new automated test, `AppointmentFlowIntegrationTest.loginResponseTokenWorksAsBearerAuthorization`, logs in, reads the token out of the JSON body with **no cookie involved at all**, and successfully calls a protected endpoint using only the bearer header - proving the mechanism live, not just asserting it exists.
- `docs/ASSIGNMENT_REPORT.md` §3.6 was rewritten to explain the full JWT authorization flow (signing, claims, expiry, validation, both delivery mechanisms) as the report's central security narrative; `docs/SETUP.md` gained a worked curl example and a Swagger "Authorize" walkthrough; four Swagger screenshots were recaptured showing the padlock icons, the `token` field in the schema, and a live executed login response containing a real signed JWT.

The automated suite grew from 25 to **26 tests, all passing** (see §3 below); the corresponding test-count references were updated consistently across `TEST_PLAN.md`, `TEST_CASES.md`, `SETUP.md`, `ASSIGNMENT_REPORT.md`, `testing/evidence/`, and the tests-passing screenshot was recaptured.

2. Task-by-task confirmation against the brief

Task A — System design with UML diagrams (20 marks) — Complete

Requirement	Evidence
Use Case diagram, correct actors/use cases/associations	<code>diagrams/use-case-diagram.mmd / .png</code> — 2 actors (Staff, Administrator via generalisation), all 6 brief functionalities + admin-only use cases
<code><<include>></code> / <code><<extend>></code> used accurately, with reasoning	Same diagram — 2 genuine <code><<include>></code> , 3 genuine <code><<extend>></code> with fire-condition notes (verified present in the <code>.mmd</code> source)
Class diagram, private/public modifiers, correct relationships	<code>diagrams/class-diagram.mmd / .png</code> — all 7 entities + 5 pattern classes; real <code>o--</code> (aggregation), <code>*--</code> (composition), <code>'<</code>

Requirement	Evidence
~3 Sequence diagrams	diagrams/sequence-01-login , sequence-02-register-appointment , sequence-03-generate-bill (.mmd / .png)
Assumptions documented	diagrams/README.md §"Design decisions & assumptions" (7 documented assumptions with reasoning); report §1.1, §2
Critical evaluation of the design	docs/ASSIGNMENT_REPORT.md §2.5
Bonus, beyond the brief	ER diagram (er-diagram) matching database/schema.sql exactly; program Flowchart (flowchart) tracing the menu-driven control flow

Task B — Interactive system, design patterns & architecture (40 marks) — Complete

Requirement	Evidence
(i) Distributed application with web services	3-tier: static frontend ↔ Spring Boot REST API (/api/** , 10 controllers, 29 endpoints) ↔ MySQL. Report §3.1
(ii) Appropriate design patterns	5 GoF patterns + DAO/Repository, each with problem/implementation/critical evaluation: Singleton (AppointmentNumberGenerator , AppConfigManager), Builder (AppointmentBuilder), Factory Method (BillFactory), Strategy (PricingStrategy family), Observer (AppointmentObserver / AppointmentEventPublisher). Report §3.3
(iii) Proper database	database/schema.sql : 7 tables, FKs, CHECK constraints, 2 triggers, 1 stored procedure, 1 function, 2 views — all genuinely called from Java (JdbcTemplate in ReportServiceImpl), not decorative. Report §3.4
Validation mechanisms	Layered: client-side, Jakarta Bean Validation, business-rule layer, database layer. Report §3.5
Reports for decision-making	Dashboard KPIs, daily revenue (stored procedure), dentist utilisation (view) — ReportController / ReportServiceImpl
Sophisticated UI, separate windows	13-page frontend (frontend/pages/*.html), Bootstrap 5, no CDN. Report §3.7
Complex functionality (alerts)	Observer-driven simulated Email/SMS/Audit notifications on every appointment lifecycle event. Report §3.3.5
Sessions/cookies	HttpOnly dc_token cookie. Report §3.6
Explicitly added this session	JWT bearer-token authorization (§1.2 above), now the report's primary security narrative

Task C — Testing (20 marks) — Complete

Requirement	Evidence
Test rationale + TDD explanation	testing/TEST_PLAN.md §3 — full red→green→refactor narrative for the pricing/Strategy logic

Requirement	Evidence
Test data devised/derived	<code>testing/TEST_PLAN.md</code> §5 — boundary values (4 vs. 5 prior visits, Fri/Sat/Sun dates, same-day past time)
Test plan produced and applied	<code>testing/TEST_PLAN.md</code> (full document)
Test classes created	9 JUnit 5 classes under <code>backend/src/test/java/...</code>
Tests carried out, documented	<code>testing/TEST_CASES.md</code> — 40+ traceable rows (positive/negative/boundary/validation/API/DB/integration), each mapped to its exact automated test method or manual Postman request
Demonstrate code passes all tests (screen-grab)	<code>testing/screenshots/14-tests-passing.png</code> — real <code>./mvnw test</code> output, Tests run: 26, Failures: 0, Errors: 0, BUILD SUCCESS
Test automation	<code>./mvnw test</code> (one command) + <code>.github/workflows/ci.yml</code> (runs on every push)
Evaluate success/failure incl. lessons learned	<code>docs/ASSIGNMENT_REPORT.md</code> §4.4 — three real, honestly-recorded lessons (2 proactively applied from prior work, 1 genuinely hit and fixed during this build)
Traceability	<code>testing/TEST_CASES.md</code> 's "Automated In" column; report §4.5

Current automated total: 26 tests, 0 failures, 0 errors (`testing/evidence/*.txt`).

Task D — Git, GitHub & version control (20 marks) — Complete

Requirement	Evidence
Public Git/GitHub repository	https://github.com/TharusanK23/sunrise-dental-clinic (public)
Several versions, updated with new features applied to the initial upload	21 milestone commits on <code>dev</code> (scaffold → entities → patterns → security → services/controllers → schema → tests → docs → diagrams → frontend → screenshots/exports → this session's JWT/credential update)
Version control techniques demonstrated	<code>docs/GIT_WORKFLOW.md</code> — <code>.gitignore</code> , descriptive milestone commits, a dedicated <code>dev</code> (active) branch separate from <code>release</code> (stable), the <code>mvnw</code> executable-bit fix applied proactively before it could break CI
Workflow (CI/CD) demonstrated, deployment of changes	<code>.github/workflows/ci.yml</code> — builds + runs the full JUnit suite on every push to <code>dev</code> / <code>release</code> / <code>feature/**</code> and on PRs; every run so far has gone green on the first attempt , including the run just triggered by this session's changes
Latest version deployed and demonstrated in the documentation	Both <code>dev</code> and <code>release</code> are at the same latest commit, both pushed, both green; <code>docs/ASSIGNMENT_REPORT.md</code> §5 and this document reference the live repository directly

3. Where everything lives

Deliverable	Location
Assignment report (source)	<code>docs/ASSIGNMENT_REPORT.md</code>
Assignment report (submission-ready exports)	<code>docs/ASSIGNMENT_REPORT.pdf</code> , <code>docs/ASSIGNMENT_REPORT.docx</code> — A4, margins 1.5in/1in, 1.5 line spacing, Times New Roman, 14pt bold headings, 12pt body, page numbers bottom-right, per the brief's exact format spec
Diagrams	<code>diagrams/</code> (7 diagrams: Use Case, Class, 3× Sequence, ER, Flowchart — Mermaid source + rendered PNGs)
Database schema	<code>database/schema.sql</code>
Backend source	<code>backend/src/main/java/com/sunrise/dentalclinic/</code>
Automated tests	<code>backend/src/test/java/com/sunrise/dentalclinic/</code> (9 classes, 26 methods)
Test plan / test case matrix / evidence	<code>testing/TEST_PLAN.md</code> , <code>testing/TEST_CASES.md</code> , <code>testing/evidence/</code>
Postman collection	<code>testing/postman/SunriseDentalClinic.postman_collection.json</code>
Frontend	<code>frontend/</code> (13 pages)
Setup guide	<code>docs/SETUP.md</code>
Git workflow	<code>docs/GIT_WORKFLOW.md</code>
CI pipeline	<code>.github/workflows/ci.yml</code>

4. Running the system right now

What	URL
Frontend	http://localhost/sunrise-dental-clinic-client/
Backend API	http://localhost:8082/api
Swagger UI	http://localhost:8082/swagger-ui/index.html

Account	Username	Password	Role
Administrator	<code>admin</code>	<code>admin123</code>	ADMIN
Staff	<code>kirisha</code>	<code>Kirisha@123</code>	STAFF

5. Overall status

All four tasks in the brief — A (UML design), B (interactive system, design patterns, architecture), C (testing), and D (Git/GitHub) — are complete, built to the Excellent-band criteria with evidence for every row of the marking table (full mapping in `docs/ASSIGNMENT_REPORT.md` §6), plus the two specific changes requested in this session (credential rotation and explicit JWT bearer authorization) fully implemented, tested, documented, and pushed to both the `dev` and `release` branches with a green CI run.